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Dynamic governance interactions: Evolutionary effects of state responses to non-state certification programs

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Abstract
Research has recognized that states enable or constrain private governance initiatives, but we still know too little about the interactions between private and public authority in the governance of various social and environmental problems. This article examines how states have responded to the emergence of forest and fisheries certification programs, and how state responses have influenced the subsequent development of these programs. It is argued that historical and structural differences in the management of forest and fisheries have resulted in divergent state responses to certification programs, but that both trajectories of interaction have led to a strengthening of the non-state program. The article draws upon these cases to inductively identify types of interaction between state policies and non-state certification programs, the causal mechanisms that shed light on interaction dynamics, and the conditions under which state involvement is likely to result in either strengthening or weakening of non-state programs.

Keywords: co-regulation, fisheries certification, forest certification, governance interactions, private authority.

1. Introduction

Non-state certification programs have emerged as a vibrant source of "transnational business governance" (Eberlein *et al.* 2014) across policy sectors and industries, including forestry, fisheries, aquaculture, organic agriculture, coffee production, palm oil production, and parks management. These programs typically establish environmental performance standards and standards for socially responsible production. They go beyond voluntary codes of conduct and self-regulatory modes of governing, by requiring independent verification of compliance with standards. They also constitute governing arenas in which a wide range of stakeholders interact and agree upon rules and governance mechanisms (Bernstein & Cashore 2007).

Scholars have recognized that governments enable or constrain private governance initiatives, but the dynamic interactions between private and public authority in the governance of various social and environmental problems have remained an understudied area of contemporary global governance (Vogel 2010; Eberlein *et al.* 2014). This article examines public–private governance interactions through a comparison of forest and fisheries certification programs. These programs are interesting to study because they arguably represent the most advanced cases of non-state governance dynamics in global environmental politics (Cashore 2002; Gulbrandsen

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2009). They are interesting to compare because of differences in the management of forests and of fisheries (Gulbrandsen 2010). Forests are national resources managed by domestic authorities, companies with logging concessions, and forest owners. In contrast, marine fish stocks are common pool resources managed by governments through international, regional, and domestic management regimes. While efforts to establish a binding global forest convention have met with little success, there is a long history of international governance in the fisheries sector and a wide range of legally binding rules, centered in recent decades on the 1982 Law of the Sea Convention.[1]

This article argues that historical and structural differences in the management of forests and fisheries have led to different state responses to certification programs in the two sectors, but that both trajectories of interaction have resulted in a strengthening of the non-state programs. I draw upon these cases in inductively identifying pathways of interaction between state-based regulation and non-state certification programs and the causal mechanisms that shed light on the dynamics of interaction. I also identify conditions under which state involvement is likely to result in either strengthening or weakening of non-state programs. Future research should examine whether these conditions are specific to the forestry and fisheries cases or are applicable to other cases as well. The purpose of this inductive paired comparison is to identify mechanisms of governance interactions and generate hypotheses that can be examined for a larger number of cases in future research.

This article proceeds as follows: Section 2 reviews work on the influence of states on non-state certification programs and explains the exploratory, inductive approach adopted in this study. Section 3 examines state responses to forest and fisheries certification programs, and section 4 analyzes the evolutionary effects of these responses. Based on the investigation of the forestry and fisheries cases, section 5 identifies processes, mechanisms, and hypothesized effects of interaction that can guide future research. Section 6 concludes briefly.

2. The role of states in non-state certification programs

There is now a range of voluntary standards and programs across issue areas, supplementing and sometimes supplanting traditional government regulations (see e.g. Cashore 2002; Bartley 2007; Bernstein & Cashore 2007; Potoski & Prakash 2009). While many voluntary environmental programs have been sponsored by state regulators (Coglianese & Nash 2009; Prakash & Potoski 2012), the focus of this article is on certification programs that have been created with little or no involvement from governments. Indeed, in a much-cited volume on "private authority" in fields from technical standards to credit rating, Cutler *et al.* (1999, p. 16) maintain that non-state actors are "increasingly engaged in authoritative decision-making that was previously the prerogative of sovereign states." As traditional boycott campaigns often failed to encourage companies to opt for more sustainable production practices, the creation of non-state certification programs was, in many cases, a non-governmental organization (NGO) effort aimed at finding new ways to influence corporate conduct. Other certification programs emerged as a result of the efforts of industry and business to compete with or complement NGO-sponsored schemes (Cashore *et al.* 2004; Gulbrandsen 2010). The main claim of much of the literature on private authority is not that states do not contribute to the governance processes, but that private regulatory programs do not derive rulemaking authority from governments (Cashore 2002; Bartley 2007).

Although research has recognized that states may influence private governance efforts in several ways, we still know too little about the dynamic interactions between private and public

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authority in the governance of various social and environmental problems. How can states influence non-state programs? It is useful to differentiate among the regulatory contributions that actor groups make at the different stages of the regulatory process, depending on their "competencies" (Abbott & Snidal 2009).[2] Abbott and Snidal (2009) identify five stages of the regulatory process: agenda-setting, negotiation of standards, implementation, monitoring, and enforcement. For this analysis, I reduce these five stages to three – agenda-setting/negotiation, implementation, and monitoring/enforcement – and identify the regulatory contributions states can make at each stage.

Beginning with agenda-setting and negotiation, governments can participate in these processes by providing expertise and technical advice, as well administrative or financial support (Boström 2003; Meidinger 2006; Tollefson *et al.* 2008; Lister 2011). Government may influence agenda-setting and negotiation both through direct regulation and "the shadow of hierarchy." After the 1984 Bhopal disaster in India, for instance, there has been strong legal incentive for firms in the chemical industry to join voluntary programs; this can demonstrate their due diligence and shield them from punitive fines in the event of an industrial accident. The Responsible Care program of the chemical industry emerged and diffused as a direct result of this sort of pressure (Prakash 2000). Governments can also limit the rulemaking authority of specific certification programs by supporting the creation of competing programs or enacting regulations that restrict their discretion (Gulbrandsen 2010; Gale & Haward 2011). Similarly, existing rules and regulations can facilitate or constrain non-state standard-setting processes. The key distinction is whether governments use their competencies to *restrict* or *enhance* the rulemaking authority of non-state certification initiatives (Cashore *et al.* 2004, p. 21).

Concerning implementation, Abbott and Snidal (2009, p. 66) argue that while the importance of states is relatively high at the agenda-setting/negotiation stage, it is relatively low at the implementation stage, as "even strong states lack the authority and expertise to implement standards within firms." But governments can, and often do, act as clients for non-state programs through the certification of state-controlled or -owned operations (Auld *et al.* 2008; Gulbrandsen 2010). Indeed, by certifying such operations, states indirectly take on responsibility for implementation. Governments can also require state-owned companies to adopt standards to help streamline the collection and interpretation of data on social and environmental performance, and they can provide financial support or other resources to companies or groups seeking certification (Boström 2003; Tollefson *et al.* 2008; Lister 2011). In addition to such supply-chain support, governments can be central to the uptake of certification on the demand side (Auld *et al.* 2008). The most obvious type of government demand-side support is found with public procurement policies that stipulate the purchase of certified products. This form of support applies more readily to some resources than others: for example, procurement policies are likely to play a significant role in the forestry sector, but not the fisheries sector, because the state buys many products from the former sector, but not the latter. Whereas governments could facilitate market acceptance of certification, they could also impede the uptake of certification by rejecting particular schemes or labels. If governments favor one certification scheme over another in public procurement policies, it would be a strong signal to groups considering various options. Again the key question is whether governments use their competencies to approve or reject particular certification schemes.

Finally, states may play a key role in the monitoring and enforcement of standards (Abbott & Snidal 2009, p. 69). In the natural resource programs discussed here, states are not directly involved in the monitoring and enforcement of standards. These programs rely on regular third-party audits by independent certification bodies (certifiers) for enforcement. All the same,

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they require effective regulation and enforcement of contract law, property rights, planning rules, and the like, in order to function (Meidinger 2006), and governments can have significant impact on the work of third-party auditors. Fisheries management auditors depend upon reliable fish-stocks data from public authorities for compliance assessments. They also depend upon effective government enforcement of fisheries regulations, and sometimes upon expert advice from public agencies. Forest management auditors are less dependent on public authorities for assessments, but they cannot award a certificate unless there is public enforcement of forestry laws and land tenure rights. In sum, states can facilitate or impede non-state certification at different stages of the regulatory process, as summarized in Table 1. This table does not provide an exhaustive overview of the regulatory contributions of governments, but the actions described here are believed to be the most important ones for the cases analyzed in this study.

Given that little is known about the processes, outcomes, and underlying mechanisms of public–private interaction dynamics, an exploratory, inductive approach seemed the appropriate research design for this study. I proceed in three analytical steps to investigate interaction dynamics. First, I examine how states have responded to the emergence of certification in the forest and fisheries sectors. A useful indicator of state engagement is incorporation into state policies. In recent years, several countries have developed timber procurement policies and recognized certification under credible schemes as assurance of legally and sustainably sourced timber. Hence, for the forestry case, I investigate policy documents, official assessments, and documentation from each country with timber procurement policies. As noted, investigating procurement policies is less relevant in the case of fisheries. Here I focus instead on incorporation of certification into seafood policy strategies and how states have otherwise acted to enhance or restrict the rulemaking authority of certification programs in the fisheries sector.

Second, I examine the evolutionary effects of state responses to certification programs by investigating program uptake and the development of certification standards, rules, and procedures. I draw on policy documents from forest and fisheries certification programs and primary research on these programs conducted over several years, including 27 semi-structured interviews with representatives of the programs, environmental NGOs, industry associations, and government agencies. The interviews were not conducted specifically for the purpose of investigating governance interactions, but this emerged as a key topic in many interviews.

Finally, from the empirical examination of the forestry and fisheries cases, I identify types and pathways of public–private governance interactions. I discuss underlying mechanisms of interaction dynamics and generate hypotheses on how such dynamics may influence processes

Table 1 Regulatory contributions of governments at different stages of the regulatory process

	Agenda-setting and negotiation	Implementation	Monitoring and enforcement
Government contributions	Provide expert and technical advice Provide financial or administrative support Set regulatory frameworks Enact rules that regulate activities of certification programs	Act as client for certification Provide resources to clients for certification Enact public procurement policies	Provide data for assessments Provide expert advice Enforce public rules and regulations

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and outcomes. With a limited number of cases, the value of this research lies not in offering a definite test of hypotheses, but more in identifying processes and mechanisms of interaction that can guide further research.

3. State responses to the emergence of certification

3.1. Forest certification

Forest certification emerged in response to growing concern about global forest degradation and failed intergovernmental efforts to develop a legally binding agreement on forests. In 1993, the World Wildlife Fund (WWF) officially founded the Forest Stewardship Council (FSC) in partnership with other NGOs and social groups, as well as some retailers, forest companies, and professional certifiers. Backed by governments in many producer countries, forest industry and landowner associations responded to the emergence of FSC by creating "industry-friendly" certification programs with more flexible and discretionary standards. These national-level programs eventually formed an umbrella organization, since 2003 known as the Programme for the Endorsement of Forest Certification (PEFC). Scholars have shown that governments facilitated the development of PEFC or FSC certification in many jurisdictions (see Boström 2003; Tollefson *et al.* 2008; Gale & Haward 2011; Lister 2011), but less attention has been paid to the development and effects of public procurement policies. This section provides a systematic examination of the development of such policies.

The European Parliament has, on several occasions, called for European Union (EU) restrictions on the import of tropical timber, but such restrictions have never been enacted for fear of violating multilateral trade rules. However, the EU has developed an action plan to counter illegal logging, known as the Forest Law Enforcement, Governance and Trade (FLEGT) action plan (see also Overdevest & Zeitlin 2014). This action plan, approved in Council Conclusions in October 2003, requires the EU to develop Voluntary Partnership Agreements with producer countries on timber licensing. Producer countries entering into such agreements with the EU commit to exporting only legally logged timber to the EU (European Commission 2003). Other key elements in the FLEGT action plan include the development of member-state procurement policies to purchase timber from legal sources; the promotion of private-sector initiatives, including codes of conduct and independent monitoring; and the exercise of due diligence by export credit agencies and financial institutions that fund logging in producer countries (European Commission 2003). In 2008, the European Commission proposed a new, legally binding EU Timber Regulation that recognizes certification as a tool in the risk-assessment procedure when timber is imported into the EU. This timber regulation, adopted by the European Parliament and the Council in 2010, will enter into force in 2013.

According to the EU's policy on green procurement, legality should be a minimum requirement for wood-based products (European Commission 2008), but this policy is not legally binding on the member states. Unlike trade, public procurement is a member-state competence, which leaves the final decision to each individual member state. Six EU member states – Belgium,[3] Denmark,[4] France, Germany,[5] the Netherlands,[6] and the UK[7] – have adopted procurement policies on purchasing timber from legal and sustainable sources. Outside the EU, such policies have been adopted by Japan,[8] New Zealand, and Norway.[9] Several other countries, mostly EU member states, are considering developing timber procurement policies. Although the US in 2008 became the first state to legislate against the handling of timber which is illegal according to the laws of the country the timber originated from (see Cashore & Stone 2014), it has thus far not been willing to develop procurement policies for timber.

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Examination of the six EU member states and the three other states with procurement policies shows that although critical differences remain, there has been convergence in such policies over time.[10] First, some countries, notably the UK, the Netherlands, and Denmark, have developed their own criteria for legally and sustainably sourced timber, and then assessed the degree to which certification programs meet those criteria. The UK has established a Central Point of Expertise on Timber (CPET) to assess certification schemes and advise the government on timber procurement. Similarly, the Netherlands has established a Timber Procurement Assessment Committee (TPAC) to evaluate both international and national-level certification schemes. Most other countries have adopted simpler systems, deciding that certification under certain schemes is proof of legality and sustainability. Most countries, including the UK, the Netherlands, and Denmark, have also decided that public contractors may opt to use some other credible type of assurance that the source is legal and sustainable. In the UK, for example, FLEGT-licensed timber will be accepted until 2015; thereafter, proof of sustainable timber will be required.

Second, most countries accept FSC as evidence of legal and sustainable timber from all regions and producer countries. When European governments began to develop public procurement guidelines in the early 2000s, the FSC quickly became the program of choice. These governments saw forest certification as a way to circumvent multilateral trade rules that prevented them from imposing import restrictions on tropical timber. By contrast, Norway has since 2007 banned the use of all tropical timber, including FSC-certified timber, in public-sector buildings and other construction works (Norwegian Ministry of the Environment 2007). Although this policy might be in contravention of multilateral trade rules, it has not yet been challenged by any producer country.

Third, although several countries initially signaled a preference for FSC-certified wood, most countries have eventually accepted PEFC as a credible certificate too. For example, while Denmark in 2003 declared that the FSC was the only certification scheme that provided adequate assurance of legally *and* sustainably produced tropical wood (Danish Ministry of the Environment 2003), the Danish government in 2008 accepted both FSC and PEFC as proof of legal and sustainable wood. Similarly, PEFC was not accepted as evidence of sustainable wood in the first assessment of certification schemes by the UK government (CPET 2004), but by the time the government implemented preferential treatment, PEFC had improved its standards sufficiently to meet the requirements for both legality and sustainability. However, PEFC is of a different nature than FSC: whereas FSC is a global certification scheme with one set of principles and criteria, PEFC is a mutual recognition framework that endorses national-level certification schemes on the basis of certain minimum requirements. Given the significant differences among national member schemes, NGOs have repeatedly urged countries to assess each national-level scheme and accept only those that meet legality and sustainability criteria (see FERN 2009). Because such assessments would require considerable effort, most countries have adopted a less elaborate system, deciding that PEFC as an umbrella scheme meets public procurement criteria. The exception is the Netherlands, where TPAC has assessed and accepted PEFC International and five member schemes – PEFC Belgium, Germany, Finland, Sweden, and Austria. These five national schemes supply about 90 per cent of the PEFC timber on the Dutch market. Norway is an outlier too; as with FSC, Norway does not accept PEFC-certified tropical timber in public-sector buildings and other construction works.

Fourth, some countries have included social issues in their timber-procurement policies. Denmark, the Netherlands, and Belgium were frontrunners in incorporating social requirements in public procurement contracts. Since 2003, Denmark has included various social criteria in its

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definition of sustainable forest management, relating to issues such as land tenure, rights of indigenous peoples, workers' rights, and community relations (Danish Ministry of the Environment 2003). Similarly, the Dutch and Belgian policies have included social criteria as an integral element of sustainable forest management. However, the inclusion in public procurement policies of social requirements over and above those legislated for in the producer country itself remains controversial. The UK used to exclude social issues, which meant that social requirements above those legislated for in the producer country could not be included in public procurement contracts (Brack 2005). Since April 2010, however, all government procurement contracts for timber products in the UK must include social criteria as contract conditions (CPET 2010). Social issues recognized by Denmark, the Netherlands, Belgium, and the UK include rights of indigenous peoples, rights of local communities, land tenure, recognition of customary rights, workers' rights, health and safety, fair prices (only the Netherlands), multiple functions of forests (only the Netherlands), participation, dispute resolution, and law enforcement. Although NGOs have pushed hard for the adoption of social criteria as contract conditions in timber procurements policies, Germany, France, Japan, and New Zealand have not recognized social issues in their timber procurement policies.

Fifth, like EU-member countries and Norway, Japan has developed public procurement policies closely linked to certification schemes, but the policies are more discretionary and flexible than the public procurement policies in Europe. At the G8 Gleneagles Summit in 2005, the UK government used its presidency to strengthen demand-side measures on illegal logging. Of the non-EU members of the G8, Japan was the only government to agree to develop a policy of public procurement of timber from legal and sustainable sources (Humphreys 2006, p. 160). Japan's policy was adopted in 2006 and sets out the modalities for verifying legality and sustainability. Unlike some European governments, Japan has not carried out assessments of certification schemes, but accepts FSC, PEFC, and all other certification schemes as evidence of legality and sustainability. Japan even accepts self-declarations under industry codes of conduct or company-determined methods.

To conclude, most countries with timber procurements policies consider FSC and PEFC certificates to be evidence of the legality and sustainability of forest products. Hence, timber procurement policies in Europe, Japan, and New Zealand are closely linked to forest certification schemes.

3.2. Fisheries certification

Despite important advances in intergovernmental fisheries governance in recent years, three-fourth of the world's fisheries are at or beyond "full exploitation" (FAO 2010), indicating that overall fishing pressure has remained heavy. In response to increasing concerns over the inability or unwillingness of governments to resolve the challenges of fisheries management – and inspired by the FSC's success – the WWF exported the certification and labeling idea to the fisheries sector. The Marine Stewardship Council (MSC) was formally established in 1997 as a non-profit organization. Unlike the FSC standard, the MSC standard builds on international agreements and guidelines, particularly the standards and principles in the 1995 FAO Code of Conduct for Responsible Fisheries. The MSC is therefore embedded in a regulatory field that strongly influenced state responses to the program and the subsequent evolution of its standards and procedures.

Unlike many standardization bodies, the MSC allocates no preferred position to governments, which are treated like other stakeholders such as conservation organizations, fishing companies, industry associations, and seafood buyers. Given the long history of

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intergovernmental fisheries governance, some governments were deeply skeptical of the MSC from its inception, questioning the right of non-governmental bodies to govern common-pool resources, such as fish stocks. Seeing the MSC as an attempt to create a non-state management regime beyond national jurisdiction, these governments argued that non-governmental actors had neither the necessary experience nor the mandate to govern fisheries. The Nordic Council of Ministers quickly became a central proponent of an FAO-endorsed labeling scheme, but neither through meetings in the FAO Committee of Fisheries (COFI) nor through consultation was agreement reached about how FAO should respond to the emergence of fisheries certification (see Gulbrandsen 2009, pp. 656–657). In 2003, at another COFI meeting, governments agreed to develop *voluntary guidelines* for the eco-labeling of fish and fisheries products from wild-capture fisheries. A set of such labeling guidelines was subsequently drafted during a series of FAO expert and technical consultations. These non-binding guidelines, issued by FAO in 2005, stated that labeling programs should include objective third-party fisheries assessments using scientific evidence; transparent processes with extensive stakeholder consultation and opportunities for complaints and rules for adjudication; and standards based on the sustainability of target species, ecosystems, and management practices (FAO 2005). Although the guidelines fell short of prescribing mandatory requirements for the use of eco-labels, they represented a step toward greater government influence over non-state labeling schemes (Gulbrandsen 2009).

The MSC responded swiftly by welcoming the issuance of FAO guidelines, saying that this represented "a significant endorsement of eco-labeling as a tool to achieve the sustainable management of fisheries" (MSC 2005). In order to comply fully with the guidelines, the program had to separate the standard-setting and accreditation functions. The MSC thus outsourced accreditation decisions to Accreditation Services International – an independent organization that also accredits certifiers for the FSC. Similar to the FSC, MSC's approach to accreditation has therefore evolved from an in-house process to one controlled by a separate organization. Further, the MSC was obliged to make its procedure for handling objections to fisheries assessments independent of the certification program. The MSC reported both changes had been implemented by September 2006 (MSC 2006).

Despite the initial mistrust of the MSC, many governments with large fishing industries have become increasingly supportive of the program. Rather than seeing the MSC as a potential threat to states' sovereign authority to manage fisheries, these governments now view it as a helpful supplement to domestic, regional, and international management regimes. Government management authorities have in many cases provided financial support for MSC certification or have even been the clients for certification (Gulbrandsen 2009; Gale & Haward 2011; Foley 2012). For instance, the Alaska Department of Fish and Game was the client for the certification of the Alaska salmon fishery, which re-entered assessment for its third, five-year MSC certification in July 2012. In Canada, where the government had been critical to the MSC, the federal Department of Fisheries and Oceans in 2007 made certification and eco-labeling a cornerstone of its Ocean to Plate policy strategy. This strategy was accompanied by presentations and government documents more or less officially endorsing the MSC (Foley 2012). A similar pattern of support is found across the Atlantic, where several European governments in the mid-2000s shifted from skepticism toward the MSC to active support. The UK, Norway, Denmark, Sweden, and the Netherlands have all made certification part of their seafood policy strategies and have provided funding for pre-assessment or full assessment of fisheries (interview with MSC policy officer, 2 July 2012).

A few government-backed certification schemes for wild-capture fisheries have emerged, notably in Iceland, Japan, and the US state of Alaska. Dissatisfied with the MSC, these territorial

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schemes used the FAO guidelines as a point of reference when they started developing their standards. Indeed, the FAO guidelines have become the key global "meta-governance" standard for fisheries certification schemes and political responses to them (Foley 2012). However, the MSC has continued to hold a virtual monopoly in setting widely adopted certification standards for wild-capture fisheries (Ponte 2012, p. 301) and the government-backed territorial schemes do not challenge MSC's position.

To summarize, state responses to the emergence and development of the MSC have included establishing intergovernmental guidelines for fisheries certification, providing support for MSC certification, and endorsing certification and labeling in seafood policy strategies. Notwithstanding the emergence of a few government-backed schemes, many governments with a significant stake in seafood exports have become increasingly supportive of the MSC, facilitating industry uptake of certification within their jurisdictions – much as in the case of forest certification.

4. Evolutionary effects of state responses to certification

4.1. Evolution of forest certification

We have seen that timber procurement policies are closely linked to forest certification schemes. The key question to be examined here is what these policies have meant for the development of certification. One effect of timber procurement policies has been to increase the uptake of certified products, particularly on the European market. Because there are no official statistics on public-sector timber consumption, it is difficult to estimate how much timber, out of total timber demand, is covered by procurement policies that can be shown to engage with private certification schemes. According to one estimate by the International Tropical Timber Organization (ITTO 2010), public procurement policies amount to between three per cent and 20 per cent of total timber consumption, depending on the importing country and market segment. In the EU-27 market (the 27 current EU member states), around 25 per cent of timber import is estimated to be subject to legality and sustainability verification requirements (Oliver 2009). The six EU member states with public-sector timber procurement policies account for two-thirds of total EU-27 wood-product imports from tropical saw logs and veneer logs (Oliver 2009). Only two countries – the Netherlands and the UK – have examined the impacts of public procurement policies on overall supplies. In both countries the proportion of certified timber products imported had "grown steadily" every year since the introduction of procurement policies (Brack & Buckrell 2011, p. 11). It is difficult to isolate the independent effects of public procurement, NGO pressure, and industry commitment to environmental responsibility on market uptake of certified products, but according to one estimate "it seems likely that procurement policy has had the greatest single impact" (Brack & Buckrell 2011, p. 11).

Public procurement policies not only facilitate public-sector uptake of certified products, they also help enhance the rulemaking authority of forest certification schemes. By approving specific certification schemes, governments signal that those schemes are credible governance systems on which private procurers and other buyers can rely. Evidence suggests that suppliers' preference for relatively simple supply chains magnifies this effect: timber suppliers are increasingly switching over to certified products for all of their customers, not only public-sector ones (Brack & Buckrell 2011). This switch has also helped many timber companies to spread procurement policy to local government, which is not covered by central government policy (Brack 2008a, p. 6). Procurement policies have thus had much broader effects on the timber market than simply through the direct effect of central government purchases.

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The most interesting effect of procurement policies to emerge from this analysis is the direct influence on the rules and procedures of forest certification schemes. Since the establishment of the PEFC umbrella scheme, a controversial issue has been the consistency and robustness of an arrangement that has a single universal label, but covers many national-level schemes with different standards. Unlike the case of FSC, there was no global set of principles and criteria stipulating requirements for forest operations. PEFC did have a set of basic requirements for national-level certification programs, first established in 2002 and revised many times since, but many of the specific requirements for forest operations were left to the discretion of national member programs. For these reasons, the Dutch TPAC has, as noted, not only assessed PEFC International, but also several member schemes. Similarly, CPET, established by the UK government, has examined the consistency of member schemes in assessments of PEFC International. In 2004, CPET concluded that PEFC did not meet the public procurement requirements for sustainable forest management because of unbalanced governance, inadequate public consultation during the certification process, and lack of public disclosure of auditing outcomes (CPET 2004). The American Sustainable Forestry Initiative (SFI) – since 2005 a PEFC member scheme – was not approved because the chain-of-custody certificate did not specify the amount of uncertified material used in the end product (CPET 2004).

The schemes that did not pass the test were allowed six months, beginning November 2004, to improve their standards and rules before the UK government implemented preferential treatment (ENDS 2004). In response, PEFC took up each of the CPET issues at its 2004 General Assembly and on a second especially scheduled meeting in April 2005. Subsequently, PEFC moved to adopt new conventions on balanced governance, public consultation, and transparency (PEFC 2005; Overdevest 2010, pp. 67–68). Similarly, in April 2005, the SFI adopted new chain-of-custody rules and requested that the scheme be submitted for reassessment against the UK government requirements (CPET 2005).

In August 2005, the UK government announced that PEFC and SFI had improved their standards sufficiently to meet the public procurement requirements; but they were put on probation until the end of 2005, when they would be reassessed by CPET (ENDS 2005). Greenpeace, Friends of the Earth, and other NGOs immediately attacked the government's decision, saying in a joint statement that the UK government had approved two schemes that "allow large-scale, unsustainable logging in ancient forest areas, the destruction of endangered species and the abuse of indigenous people's rights" (ENDS 2005). Notwithstanding such NGO protests, CPET's reassessment confirmed that PEFC and SFI – like FSC – could be used as evidence of legal and sustainable timber (CPET 2006).

In 2010, the PEFC General Assembly adopted a new meta-standard for sustainable forest management to improve the international system and enhance consistency across member schemes (PEFC 2010). The requirements in this standard document, which entered into force in May 2011, must be reflected in all forest management standards submitted for PEFC endorsement. Member schemes endorsed by PEFC before May 2011 must comply with the new standard by May 2013. Hence, public assessments have contributed to an upward harmonization of PEFC member schemes through the adoption of similar forest management rules and conventions on stakeholder participation, transparency, and balanced governance.

Finally, several certification schemes are developing legality assurance standards in response to the EU FLEGT and Timber Regulation, as well as member-state procurement policies (Proforest 2010b; Overdevest & Zeitlin 2014). The FSC works with government officials to ensure that its controlled wood standard (first approved in 2004, revised in 2006), developed for non-FSC certified material used in FSC Mix products, is consistent with these regulations and

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that FSC certification meets the legality requirements in the EU. At the 2011 General Assembly, the FSC membership unanimously approved a motion to strengthen the standard. A main objective for this process, targeted for completion by the end of 2012, is to "bring the Controlled Wood standard in link with EUTR [EU Timber Regulation], Lacey Act [US] and other national legality legislation."[11] Hence, while tracking systems developed in non-state certification programs have proven central in advancing international efforts to eliminate trade in illegal wood (Auld *et al.* 2010), international legality requirements are influencing certification standards and creating opportunities for expansion into new areas of verification.

To conclude, just as Overdevest (2010) argues for the importance of public comparisons in her work on ratcheting forest certification standards, procurement policies have forced the public exposure of information about certification standards, putting pressure on poorly performing programs to improve their standards. Public comparisons with FSC's higher standards have pushed PEFC International and its member schemes to increase the stringency of their standards, improve governance procedures and enhance participation from outside stakeholders in certification processes. Similar to the Trading-Up hypothesis developed by Vogel (1995), certification programs have converged upwards to satisfy public procurements requirements in "green markets" – the European timber trade market in particular.

4.2. Evolution of fisheries certification

Unlike the FSC, we have seen that the MSC standards were based upon intergovernmental agreements, particularly the 1995 FAO Code of Conduct. Despite this linkage to intergovernmental agreements, governments were not willing to give the MSC full discretion to establish a global standard for certification of wild-capture fisheries. The development of Food and Agriculture Organization (FAO) guidelines on fisheries eco-labeling, issued in 2005, was essentially an intergovernmental effort to reassert authority in assessing credible standards and certification schemes (Gulbrandsen 2009). The *effect* of these guidelines, however, has been to strengthen MSC's position as the "gold standard" for certification of wild-capture fisheries. Instead of presenting a challenge to MSC authority, the FAO guidelines have consolidated the MSC's position as the leading global certification program in the fisheries sector, making it more difficult for potential competitors to create schemes with equally strict requirements. Importantly, the three main principles of fisheries assessment identified by FAO correspond to the MSC principles – health of the fish stocks, impact of the fishery on the ecosystem, and performance of the fishery management system (MSC 2002). Moreover, the guidelines stipulate that any credible fisheries eco-labeling program must be independent, transparent, and science-based, and have objections procedures and a high degree of stakeholder involvement. As Rupert Howes, the chief executive of MSC, stated when the FAO guidelines were issued: "there is no other [certification] organization that is remotely near FAO compliance; the MSC is the only one" (interview, 23 May 2006).

The growth of the MSC following the issuance of the 2005 FAO guidelines is evident when we examine the number of certified fisheries and the total certified catch. After a slow start, the MSC experienced rapid growth during the second half of its first decade of operations. The number of certified fisheries increased from 12 in 2005 to 135 by the end of 2011 (MSC 2011). Another 136 fisheries were in some stage of full assessment and 40 to 50 more were estimated to be in confidential pre-assessment (MSC 2011). These fisheries (certified and in assessment) record annual catches of over 9 million tons of seafood – over 10 per cent of the annual global harvest of wild-capture fisheries (MSC 2011). MSC-certified fisheries represent close to 50 per cent of the whitefish market, more than 40 per cent of the wild salmon market, and about 18 per

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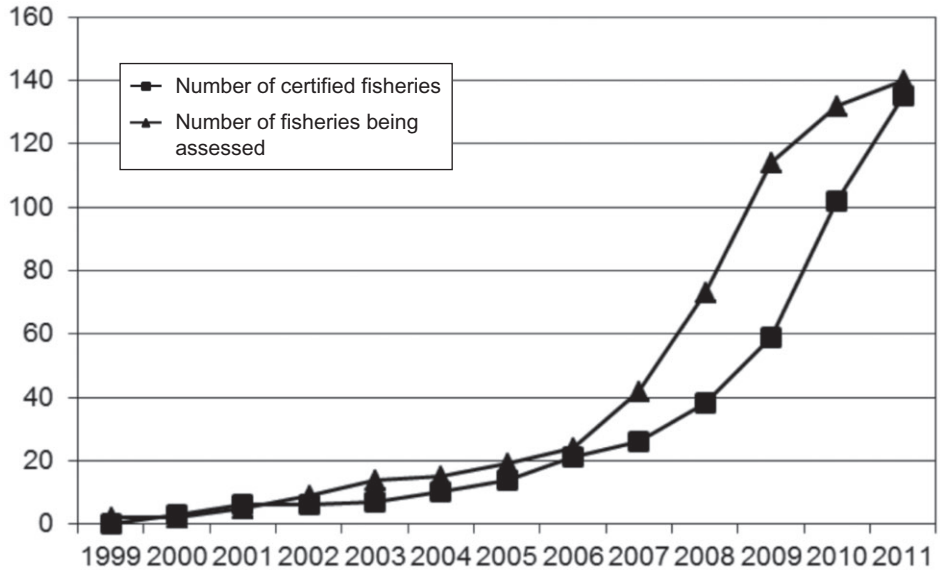

Figure 1 Number of certified fisheries and fisheries in assessment by year, 1999–2011 (as of 31 March each year).
Source: Data provided by the Marine Stewardship Council (MSC) on request (March 2012).

cent of the lobster market. For other fish species, the market share of certified products is much lower: for example, only 0.5 per cent of the tuna market is certified. Figure 1 shows the number of certified fisheries and fisheries in assessment from the first certifications in 2000 to mid-2011.

Given the many factors that have contributed to the growth of the MSC, it is difficult to isolate the independent effect of the FAO guidelines on MSC uptake. Other government policies, market demand, NGO pressure, and a growing retailer commitment to source sustainable seafood are all factors that have influenced MSC uptake (Gulbrandsen 2010). Indeed, a major breakthrough came in January 2006, less than one year after the issuance of the FAO guidelines, when Walmart – the world's biggest retailer – announced a commitment to source all of its fresh and frozen seafood supplies in North America from MSC sources within five years. According to the MSC chief executive, the Walmart commitment pressured its suppliers to become involved in the MSC, and spurred other retailers to look at their own commitments to the program (interview with Rupert Howes, 23 May 2006). Similarly, other analysts have highlighted how Walmart played an important role in raising the number of certifications in the late 2000s (e.g. Ponte 2012).

The evidence suggests, however, that the FAO guidelines in combination with other government policies and market dynamics have had a positive impact on the development and uptake of the program. First, the MSC has actively employed the FAO guidelines in efforts to enhance the credibility of the program and attract new clients for certification. It welcomed the guidelines, issued a statement of full compliance as soon as the necessary organizational and procedural changes were implemented, and has been careful to communicate that it is the only global certification program consistent with them.[12] In short, the FAO guidelines represented an endorsement of fisheries certification, enabling the MSC to attract clients that otherwise would have been skeptical to the program (interview with MSC policy officer, 2 July 2012).

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Second, the fact that the MSC could rapidly demonstrate full consistency with FAO guidelines also helped convince several governments to support the program. Recall that Nordic fisheries and their governments in the 1990s had been highly doubtful as to the MSC or any other private standards in the fisheries sector, and had called for the establishment of an FAO-endorsed labeling scheme. When such efforts failed, the Nordic governments became central proponents of FAO labeling guidelines. The subsequent establishment of the FAO guidelines should be seen as an effort to reassert authority in assessing credible private standards in the fisheries sector, not to weaken or undermine the MSC. When the MSC announced full consistency with the guidelines in 2006, the Nordic fisheries and their governments (with the exception of Iceland) eventually accepted it as a credible certification program. This acceptance also has to do with the evolution of the MSC from a minor experiment in the late 1990s to a full-fledged and elaborate certification program by the time the FAO guidelines were issued.

Third, while various aquaculture certification programs have emerged, the MSC has remained the only global certification program for wild-capture fisheries consistent with the FAO guidelines. When Walmart and other big companies began to demand MSC-certified products, many governments concerned about the market access of their seafood industries decided to encourage and support certification. Government support for the MSC has expanded in recent years, as governments with significant stakes in the fishing industry have sought to secure, maintain, and expand domestic seafood exports. It is critical to recognize, therefore, that public and private rulemaking processes are closely intertwined, and that the presence or absence of one affects the dynamics in the other.

5. Mechanisms and hypothesized effects of interaction

The evidence presented in this study shows how the interactions between private and public authority shape policy trajectories and policy outcomes. Four inductively generated hypotheses follow from this analysis of interaction dynamics between states and non-state certification programs. First, the evidence indicates that when certification programs compete for public recognition, procurement policies are more likely to put pressure on poorly performing programs to improve their standards. Public procurement policies have resulted in an upward convergence of PEFC member schemes, narrowing the gap between their approach and that of the FSC. The causal mechanism at work here is *public comparison and benchmarking*, forcing the public exposure of information about certification standards and procedures, and pressuring the poorer performers to improve their standards (Overdevest & Zeitlin 2014).

Second, the evidence suggests that a process of mutual reinforcement of state policies and a non-state program is more likely to occur when the state depends on the non-state program for implementing public regulations or attaining public policy objectives. In the case of forestry, states with timber procurement policies depend upon private certification schemes for sustainability and legality verification. In the case of fisheries, many coastal states depend on the MSC for providing sustainability assurance to export markets for seafood products. In turn, non-state programs receive legitimacy and greater uptake from incorporation into public policies. In the absence of the exclusive policymaking authority of the state, non-state programs must be granted legitimacy and rulemaking authority from relevant audiences (Cashore 2002). Incorporating private standards into public policies can provide crucial legitimacy to non-state programs. Similarly, the expertise, competencies, and "social capital" (such as access to transnational networks) of non-state programs can lend credibility to state-based regulation. Symbolic resources, such as a widely recognized logo and brand name associated with a particular

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organization (like FSC), can also bring credibility to government programs. The mechanism at work here is *mutual reinforcement of legitimacy*, which occurs when the certification scheme gains legitimacy from incorporation into public policies; the state gains the credibility, expertise, and auditing capacity of the scheme; and "each gain the symbolic resources of the other" (Eberlein *et al.* 2014).

Third, the comparison of state responses to forest and fisheries certification indicates that states are more likely to reassert authority in assessing credible standards when non-state programs emerge in densely regulated sectors. Given the exclusive authority of governments in fisheries governance and allocation decisions, many governments were initially skeptical of the MSC, questioning the right of non-state certification programs to play a role in fisheries governance. In the fisheries sector, unlike the forestry sector, there is a long history of international governance and a wide range of legally binding rules. The fact that fisheries are managed by governments, not companies or private owners, explains why governments responded to the MSC by mandating the FAO with the task of developing rules for fisheries certification and eco-labeling. The FAO guidelines have become the global "meta-regulatory standards" (Eberlein *et al.* 2014) for standard setting, certification, auditing, and accreditation in the fisheries sector. Indeed, all fisheries certification and labeling initiatives claim compliance with the FAO guidelines and use them as a reference point. The mechanism at play in this case is *coercive isomorphism* (DiMaggio & Powell 1983), which occurs when the reassertion of state authority compels private certification programs to modify standards and rules. Although states do not force certification programs to comply with the FAO guidelines, it is likely that the guidelines will continue to provide the global meta-regulatory point of reference for different fisheries certification initiatives. In this analysis, FAO is viewed primarily as an agent of member state interests; the coercive aspect of interaction originates with powerful member states and works through this intergovernmental organization to impose a set of compelling normative guidelines.

Fourth, the evidence suggests that policy learning across private and public governance efforts is more likely to occur when non-state certification programs disclose not only information about rules and procedures, but also information about outcomes and effects. The mechanism at work here is *cognitive interaction*, which occurs when information, knowledge, or ideas generated in one institution change the perceptions of decisionmakers in another (Gehring & Oberthür 2009). For example, while tracking systems developed in forest certification programs have become critical for advancing international efforts to eliminate trade in illegal wood, certification programs are developing legality assurance standards in response to international legality requirements. As highlighted by Eberlein *et al.* (2014), the information non-state schemes produce and disseminate may be among the most important mechanisms of transnational governance interaction. However, incomplete information disclosure across the field of non-state certification programs presents problems for this collective model of learning and innovation (Auld & Gulbrandsen, 2012). While procedural transparency for auditing (the specific steps an audit will involve and information on when consultation and stakeholder engagement are to occur) has become a norm across all programs in the forest and fisheries sectors, several PEFC-endorsed programs disclose incomplete information about the outcomes of audits. Without full disclosure of the outcomes and effects of non-state certification, collective learning to advance earth system governance will be difficult (Auld & Gulbrandsen, 2012). Table 2 summarizes the causal mechanisms associated with the types of interaction identified in this study, with inductively generated hypotheses on interaction dynamics that can be tested in future research.

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Table 2 Public–private interaction dynamics: hypotheses, mechanisms, and examples

Hypotheses	Mechanisms	Examples
Public procurement policies are more likely to pressure poorly performing programs to improve their standards when certification programs compete for public recognition	Public comparison and benchmarking	Upwards convergence of PEFC member schemes
Mutual reinforcement is more likely to occur when the state depends on the non-state program for implementing regulations or attaining public policy objectives	Mutual reinforcement of legitimacy	Public recognition of private certification schemes in timber procurement policies and seafood policy strategies
States are more likely to reassert authority in assessing credible standards when non-state programs emerge in densely regulated sectors	Coercive isomorphism	Development of FAO guidelines for eco-labeling of fish and fisheries products
Policy learning across private and public governance efforts is more likely to occur when non-state certification programs disclose information about outcomes and effects	Cognitive interaction	Tracking systems for legally and sustainably sourced wood

FAO, Food and Agriculture Organization; PEFC, Programme for the Endorsement of Forest Certification.

6. Conclusions

Two key insights have emerged from this study of the evolutionary effects of state responses to forest and fisheries certification programs. First, state responses to private authority vary with the historical trajectory and structural or policy features of a domain. Second, different trajectories may still lead to similar outcomes. The key explanatory factors in this study are the dynamic interactions between private and public authorities in the governance of forests and fisheries and the sequence of events. In the case of forest certification, many producer countries responded to the FSC by supporting the development of producer-dominated schemes with more flexible and discretionary standards, while many consumer countries responded by developing public procurement policies stipulating the purchase of timber approved by FSC or "equivalent" schemes. The effects of government recognition of certain certificates as assurance of timber legality and sustainability were to legitimize forest certification as a global policy instrument and to pressure the poorly performing programs to improve their standards.

In contrast, states responded to the emergence of the MSC by mandating FAO with the task of developing guidelines for fisheries certification and labeling. This can be seen as an attempt by certain governments, backed by domestic fisheries associations, to regain some control over the certification program that held a virtual monopoly in setting standards for wild-capture fisheries certification. The effect of the FAO guidelines, however, was to strengthen MSC's position as the leading global certification program for wild-capture fisheries. The FAO guidelines represented an endorsement of fisheries certification as a global policy tool to promote responsible fisheries management, much as in the case of government recognition of forest certification. This outcome can partly explain the shift among several governments, from skepticism of the MSC to active support for the program. Many governments with a significant stake in seafood exports

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have become increasingly supportive of the MSC, encouraging, enabling, and endorsing fisheries certification. As with the FSC, government support for the MSC has contributed to its impressive growth in recent years. In both cases, state responses to the emergence of certification programs have served to legitimize certification as a credible policy tool that companies and consumers can trust.

Policy outcomes in the forestry and fisheries sectors are not indicative of less government involvement, but rather of public–private governance interactions at multiple governance levels and in multiple ways. More research is needed to reveal whether the causal mechanisms and the types of interaction inductively identified here are specific to the forestry and fisheries cases or may apply more generally within and across sectors. While non-state certification programs represent a remarkable policy innovation, their evolutionary potential depends critically on how and whether they will act synergistically with government rules. The key question is how the interactions between non-state programs and states influence the performance of these programs, and, ultimately, their role as an integral part of collective efforts to manage earth systems.

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Notes

1 Full title: United Nations Convention on the Law of the Sea of 10 December 1982.
2 Abbott and Snidal (2009) identify four key competencies: independence, representativeness, expertise, and operational capacity.
3 Policy documents at http://www.gidsvoorduurzameaankopen.be/ (last accessed 1 Nov 2012).
4 Policy documents available at http://www.naturstyrelsen.dk/Naturbeskyttelse/Skov/Miljoet/vejledning/revision/Udgivelser/Udgivelser.htm (last accessed 1 Nov 2012).
5 Federal Ministry of Food, Agriculture and Consumer Protection (2010) Joint instruction on the procurement of wood products, http://www.bmelv.de/SharedDocs/Rechtsgrundlagen/EN/H/ProcurementOfWoodProducts.html and Explanatory Notes regarding the procurements of wood products as of 2 December 2010, http://www.bmelv.de/SharedDocs/Standardartikel/EN/Agriculture/forestTimberHunting/ProcurementOfWoodProductsAttachment.html (last accessed 1 Nov 2012).
6 Policy documents available at http://www.tpac.smk.nl/ (last accessed 1 Nov 2012).
7 Policy documents available at http://www.cpet.org.uk/ (last accessed 1 Nov 2012).
8 Institute for Global Environmental Strategy (2007).
9 Norwegian Ministry of the Environment (2007).
10 Except where otherwise indicated, the analysis in this section is based on policy documents from each country (with the exception of France, Japan, and New Zealand, for which secondary sources were used) and the Institute for Global Environmental Strategy (2007), Brack (2008a,b), FERN (2009), ITTO (2010), Proforest (2010a), and Brack and Buckrell (2011).
11 See http://www.fsc.org/newsroom.9.21.htm (last accessed 8 Jun 2012).
12 See, for example, http://www.msc.org/about-us/credibility/how-we-meet-best-practice; http://www.msc.org/about-us/21-facts; and http://www.msc.org/business-support/key-facts-about-msc (last accessed 1 Nov 2012).

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